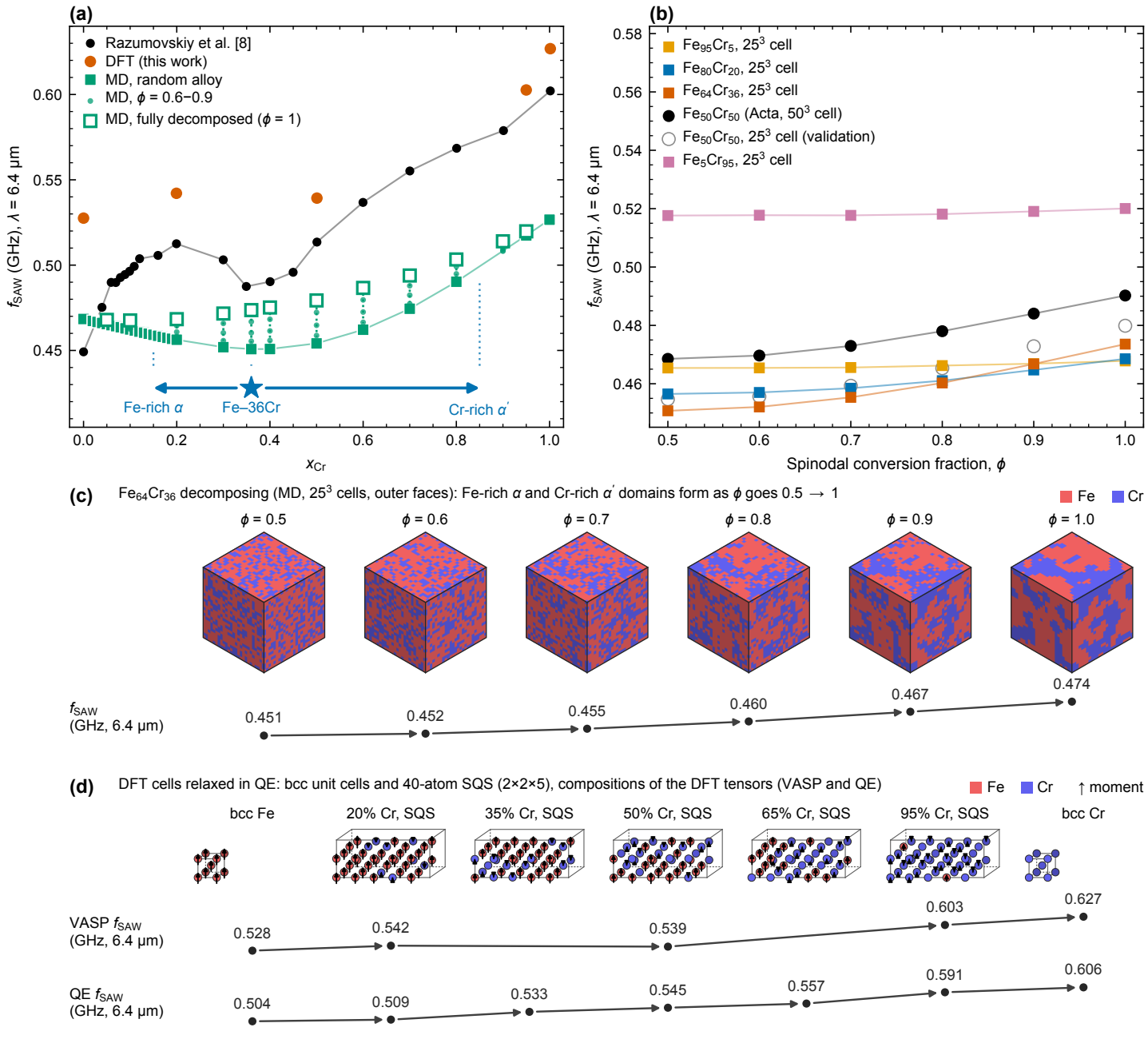

(a)
Razumovskiy et al. [8]
DFT (this work)
MD, random alloy
MD, ϕ = 0.6–0.9
MD, fully decomposed (ϕ = 1)
Fe-rich α
Fe–36Cr
Cr-rich α′
f_{SAW} (GHz), λ = 6.4 μm
x_{Cr}
(b)
$Fe_{95}Cr_5$, 25^3 cell
$Fe_{80}Cr_{20}$, 25^3 cell
$Fe_{64}Cr_{36}$, 25^3 cell
$Fe_{50}Cr_{50}$ (Acta, 50^3 cell)
$Fe_{50}Cr_{50}$, 25^3 cell (validation)
Fe_5Cr_{95}, 25^3 cell
f_{SAW} (GHz), λ = 6.4 μm
Spinodal conversion fraction, ϕ
(c)
$Fe_{64}Cr_{36}$ decomposing (MD, 25^3 cells, outer faces): Fe-rich α and Cr-rich α′ domains form as ϕ goes 0.5 → 1
Fe
Cr
ϕ = 0.5
ϕ = 0.6
ϕ = 0.7
ϕ = 0.8
ϕ = 0.9
ϕ = 1.0
f_{SAW} (GHz, 6.4 μm)
0.451
0.452
0.455
0.460
0.467
0.474
(d)
DFT cells relaxed in QE: bcc unit cells and 40-atom SQS (2×2×5), compositions of the DFT tensors (VASP and QE)
Fe
Cr
↑ moment
bcc Fe
20% Cr, SQS
35% Cr, SQS
50% Cr, SQS
65% Cr, SQS
95% Cr, SQS
bcc Cr
VASP f_{SAW} (GHz, 6.4 μm)
0.528
0.542
0.539
0.603
0.627
QE f_{SAW} (GHz, 6.4 μm)
0.504
0.509
0.533
0.545
0.557
0.591
0.606